

Producer Theory

Lecture 15: Elasticity — Demand Recap & Supply Elasticity

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2026

Recap: Lecture 14

What we covered last time:

- **Producer surplus:** $(PS = TR - VC)$ = area above supply, below price
- **Market supply:** horizontal sum of individual supply curves (kinks when firms enter)
- **Linear supply:** $(P = c + dQ)$, with $(PS = \frac{1}{2} \times Q^* \times (P^* - c))$

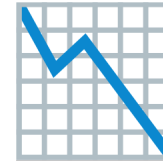
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 **Today:** How **responsive** are quantities to price changes?

Before we tackle **supply** elasticity, we need a quick recap of the **demand** elasticity concept — we ran out of time for it earlier in the course. The good news: the idea is the same on both sides!

Demand Elasticity: The Key Ideas

Price Elasticity of Demand



PRICE ELASTICITY OF DEMAND (ϵ_D)

The **percentage change in quantity demanded** resulting from a **1% change in price**:

$$\epsilon_D = \frac{\% \Delta Q_D}{\% \Delta P} = \frac{\Delta Q / Q}{\Delta P / P}$$

Since demand slopes downward ($P \uparrow \rightarrow Q \downarrow$), ϵ_D is **negative**. We often use the **absolute value** $|\epsilon_D|$.

$ \epsilon_D $	Classification	Meaning	Tourism example
> 1	Elastic	Q responds more than P	Leisure beach holidays
$= 1$	Unit elastic	Q responds exactly as P	— (theoretical midpoint)
< 1	Inelastic	Q responds less than P	Business travel
$= 0$	Perfectly inelastic	Q doesn't respond at all	Emergency medical flights
$= \infty$	Perfectly elastic	Infinite response	Identical commodity, many sellers

Calculating Demand Elasticity

Two methods (same as we'll use for supply):

1 Point elasticity (at one point):

$$\epsilon_d = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} = \frac{1}{\text{slope}} \times \frac{P}{Q}$$

Example: Demand is $(P = 50 - 2Q)$.

At $(P = 30)$, $(Q = 10)$: slope = (-2)

$$\epsilon_d = \frac{1}{-2} \times \frac{30}{10} = -1.5$$

$(|\epsilon_d| = 1.5 > 1) \rightarrow$ **elastic**

2 Arc elasticity (midpoint method):

$$\epsilon_d = \frac{Q_2 - Q_1}{(Q_1 + Q_2)/2} \div \frac{P_2 - P_1}{(P_1 + P_2)/2}$$

Example: A museum raises price from €10 to €12. Visitors fall from 1,000 to 800.

$$\epsilon_d = \frac{-200}{900} \div \frac{2}{11} = -1.22$$

$(|\epsilon_d| = 1.22 > 1) \rightarrow$ **elastic**

 Key insight for linear demand: elasticity **varies** along the curve — elastic at the top (high P, low Q), inelastic at the bottom (low P, high Q), unit elastic at the midpoint.

The Revenue Test & Determinants

The Revenue Test:

How does total revenue ($TR = P \times Q$) respond to a price increase?

Elasticity	$P \uparrow \rightarrow TR?$	Intuition
Elastic ($ \epsilon_d > 1$)	TR falls 	Q drops more than P rises
Unit elastic ($ \epsilon_d = 1$)	TR unchanged	Effects cancel
Inelastic ($ \epsilon_d < 1$)	TR rises 	Q drops less than P rises

- 👉 Hotel prices up 10%, but bookings only drop 3%? Demand is inelastic $\rightarrow$ revenue **rises**!

Five determinants of demand elasticity:

-  **Substitutes**: more substitutes $\rightarrow$ more elastic
-  **Budget share**: bigger expense $\rightarrow$ more elastic
-  **Necessity vs luxury**: luxuries more elastic
-  **Time horizon**: long run $\rightarrow$ more elastic
-  **Market definition**: narrower $\rightarrow$ more elastic

Tourism: Leisure travel is very elastic ($|\epsilon_d| \approx 2-3$), business travel is inelastic ($|\epsilon_d| \approx 0.3$). Airlines use this to price-discriminate!

Demand Elasticity: Quick Practice

Quick check — try these mentally before revealing the answers!

Q1: Algarve resort demand: $(P = 500 - 0.05Q)$. At $(P = 100)$, $(Q = 8,000)$. Is demand elastic or inelastic?

$(\epsilon_d = \frac{1}{-0.05} \times \frac{100}{8000} = -20 \times 0.0125 = -0.25) \rightarrow \text{Inelastic } (|\epsilon_d| = 0.25 < 1)$. Raising prices would **increase** revenue!

Q2: If luxury cruise demand has $(|\epsilon_d| = 2.5)$ and operators raise prices by 8%, what happens to revenue?

Elastic demand $\rightarrow$ price and revenue move in **opposite directions**. Revenue **falls**. $(\% \Delta Q \approx -20\%)$, much larger than the 8% price rise.)

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 Now let's apply the **exact same logic** to the supply side!

Price Elasticity of Supply

Definition



PRICE ELASTICITY OF SUPPLY (ϵ_s)

The **percentage change in quantity supplied** resulting from a **1% change in price**:

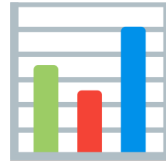
$$\epsilon_s = \frac{\% \Delta Q_s}{\% \Delta P} = \frac{\Delta Q / Q}{\Delta P / P}$$

Key differences from demand elasticity:

bulb: The formula is **identical** — the only difference is that supply slopes upward, so $\epsilon_s > 0$.

	Demand elasticity (ϵ_d)	Supply elasticity (ϵ_s)
Sign	Negative ($P \uparrow \rightarrow Q \downarrow$)	Positive ($P \uparrow \rightarrow Q \uparrow$)
We use absolute value?	Yes ($ \epsilon_d $)	No — it's already positive
Interpretation	Responsiveness of buyers	Responsiveness of sellers

Classification



Elasticity value	Classification	Meaning
$\epsilon_s > 1$	Elastic	Quantity supplied responds more than proportionally to price
$\epsilon_s = 1$	Unit elastic	Quantity supplied responds exactly proportionally
$\epsilon_s < 1$	Inelastic	Quantity supplied responds less than proportionally
$\epsilon_s = 0$	Perfectly inelastic	Quantity doesn't respond at all (vertical supply)
$\epsilon_s = \infty$	Perfectly elastic	Any quantity at one fixed price (horizontal supply)

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Tourism intuition:

- 🏨 Hotel rooms in Lisbon in the short run → **inelastic** (can't build new rooms overnight)
- 🍴 Street food vendors in a tourist area → **elastic** (easy to set up a new stall)
- 🏛️ UNESCO heritage sites → **perfectly inelastic** (supply is fixed!)

Calculating Supply Elasticity

Point and Arc Methods

1 Point elasticity:

$$\epsilon_s = \frac{1}{\text{slope}} \times \frac{P}{Q}$$

Example: Supply is $(P = 5 + 0.25Q)$. At $(P = 30)$, $(Q = 100)$:

Slope = (0.25) , so $\left(\frac{1}{\text{slope}}\right) = 4$

$$\epsilon_s = 4 \times \frac{30}{100} = 1.2$$

Elastic — a 1% price increase leads to a 1.2% increase in quantity supplied.

👉 Same methods as demand elasticity — just applied to the supply curve!

2 Arc elasticity (midpoint):

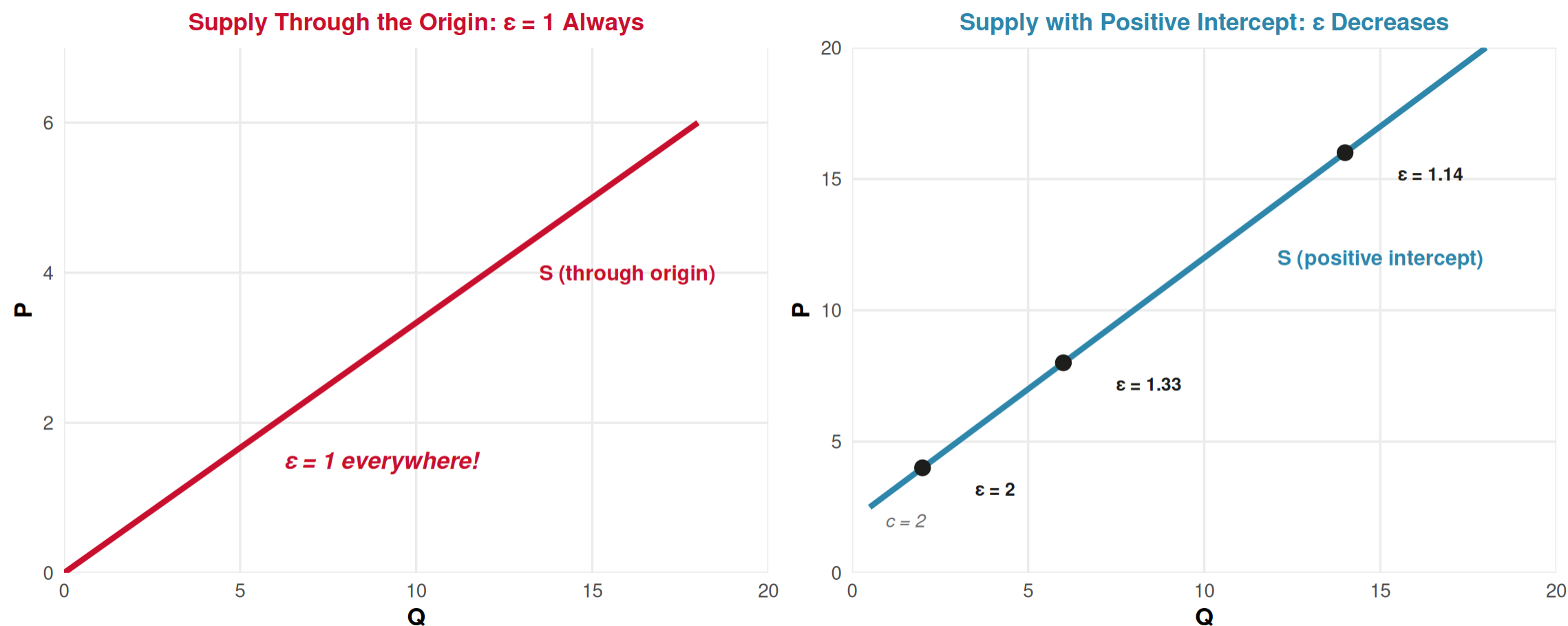
$$\epsilon_s = \frac{Q_2 - Q_1}{(Q_1 + Q_2)/2} \div \frac{P_2 - P_1}{(P_1 + P_2)/2}$$

Example: Tour operator increases tours from 40 to 50/month when price rises from €80 to €100.

$$\epsilon_s = \frac{10}{45} \div \frac{20}{90} = 0.222 \div 0.222 = 1.0$$

Unit elastic — quantity and price change by the same percentage.

Elasticity Along a Linear Supply Curve



Left: Supply through the origin $\rightarrow (\epsilon_s = 1)$ at **every** point (P and Q always change proportionally).

Right: Supply with positive intercept ($c > 0$) $\rightarrow (\epsilon_s > 1)$ near the intercept, **falling** as Q increases.

The Origin Rule & Quick Guide

Why does supply through the origin always have $(\epsilon_s = 1)$?

If $(P = dQ)$: $(\epsilon_s = \frac{1}{d} \times \frac{P}{Q} = \frac{1}{d} \times \frac{dQ}{Q} = 1)$

The (P/Q) ratio always equals (d) , which cancels perfectly with $(1/d)$.

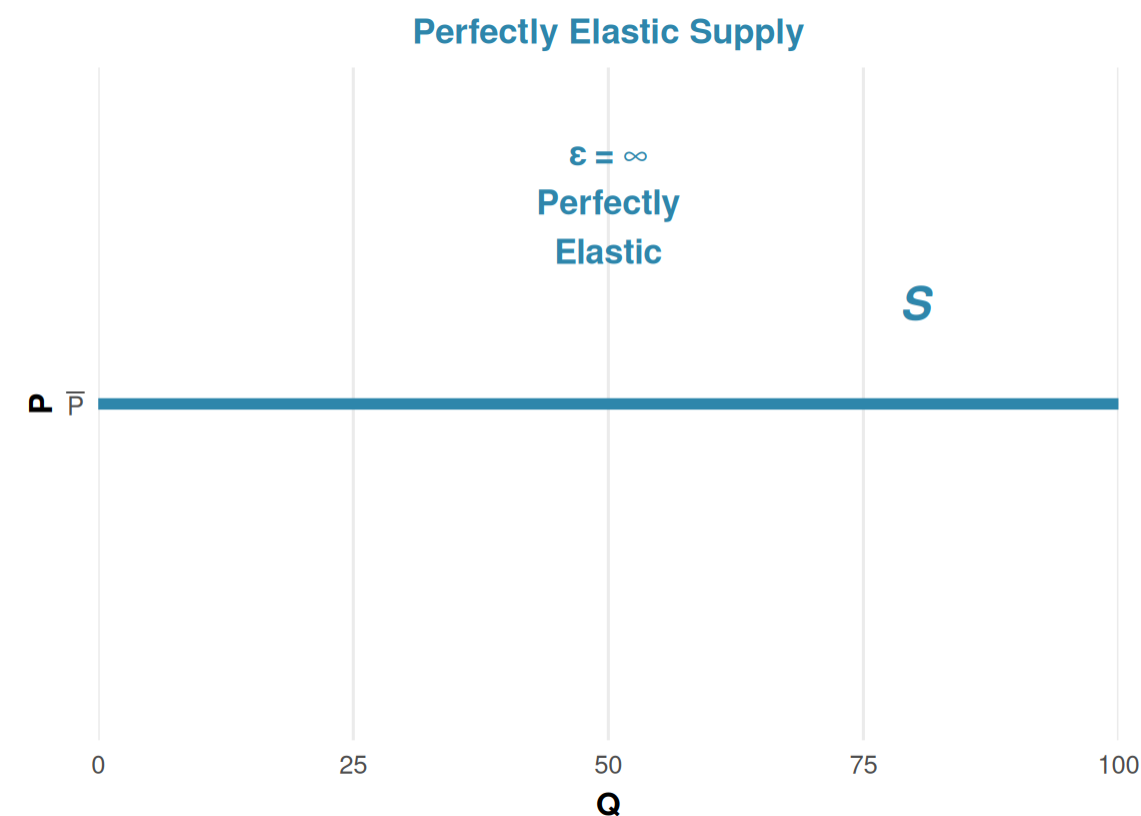
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QUICK ELASTICITY GUIDE FOR LINEAR SUPPLY $(P = C + DQ)$

Intercept	Elasticity	Shortcut formula
$(c = 0)$ (through origin)	$(\epsilon_s = 1)$ everywhere	—
$(c > 0)$ (positive intercept)	$(\epsilon_s > 1)$, falling toward 1	$(\epsilon_s = \frac{P}{P - c})$
$(c < 0)$ (hits Q-axis)	$(\epsilon_s < 1)$ everywhere	$(\epsilon_s = \frac{P}{P - c})$ (where $(c < 0)$)

 **Contrast with demand:** demand elasticity varies from $(-\infty)$ to (0) along a linear curve. Supply elasticity is more “stable.”

Extreme Cases



Perfectly inelastic ($\epsilon_s = 0$): Quantity fixed regardless of price.

Tourism: Hotel rooms in a fully built-up area, UNESCO sites, beach space.

Perfectly elastic ($\epsilon_s = \infty$): Any quantity at one price — constant MC.

Tourism: Digital products (e-tickets, online guides) — near-zero MC per copy.

Determinants of Supply Elasticity

What Makes Supply More or Less Elastic? 🤔

The textbook identifies **four key determinants** (plus time):

1 Flexibility of production factors

Can inputs be used for other things? If yes → elastic.

-  **Elastic**: Unskilled labor (waiters can switch from retail to tourism)
-  **Inelastic**: Specialized skills (brain surgeons, master sommeliers)

2 Mobility of production factors

Can inputs move between locations? If yes → elastic.

-  **Elastic**: Touring musicians, portable equipment
-  **Inelastic**: Beachfront land, historical buildings

3 Ability to produce substitute inputs

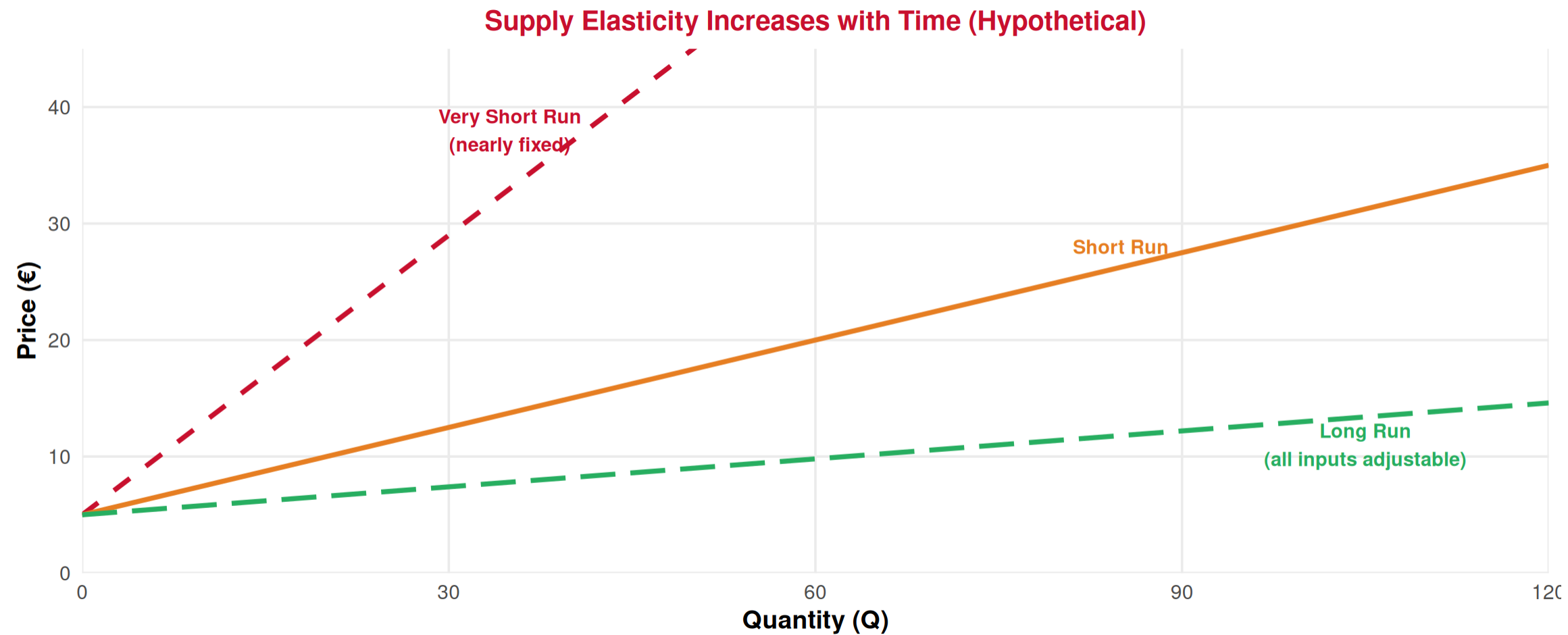
Can inputs be replicated over time? If yes → elastic.

-  **Elastic**: Train more tour guides, build more hotel rooms
-  **Inelastic**: Rare natural resources (diamond mines, volcanic hot springs)

4 Time horizon ⌚

- **Short run**: At least one input is fixed → more **inelastic**
- **Long run**: All inputs adjustable → more **elastic**
- **Very short run** (market period): supply is nearly **vertical**

Time and Supply Elasticity



The **longer the time horizon**, the more firms can adjust all inputs → supply becomes **flatter** (more elastic).

Tourism Applications

Elasticity in Tourism: Both Sides

Inelastic supply (ϵ_s small):

-  **Hotel rooms in central Lisbon**: limited space, takes years to build
-  **Museum capacity**: Jerónimos can't add rooms
-  **Beach access in Algarve**: fixed coastline

Implication: When demand rises (peak season), price rises **sharply** because supply can't respond much.

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-  **Both elasticities together** determine **who bears the burden** of taxes and price shocks! The side with **less elastic** response absorbs more of the burden.

Elastic supply (ϵ_s large):

-  **Tour buses**: can add more buses relatively quickly
-  **Food trucks at festivals**: quick to set up
-  **Online booking services**: near-zero cost to scale

Implication: When demand rises, quantity adjusts more than price — price stays relatively stable.

The Full Parallel: Demand vs Supply Elasticity

Concept	Demand Elasticity	Supply Elasticity
Formula	$\epsilon_d = \frac{\Delta Q/Q}{\Delta P/P}$	$\epsilon_s = \frac{\Delta Q/Q}{\Delta P/P}$
Sign	Negative (use $ \epsilon_d $)	Positive
Elastic	$ \epsilon_d > 1$	$\epsilon_s > 1$
Revenue test	Elastic: $P \uparrow \rightarrow TR \downarrow$	No direct revenue test
Along linear curve	Varies: $\infty \rightarrow 1 \rightarrow 0$	More stable; through origin = 1
Key determinants	Substitutes, budget share, necessity, time, market definition	Input flexibility, mobility, replicability, time
Perfectly inelastic	Vertical demand	Vertical supply
Perfectly elastic	Horizontal demand	Horizontal supply

- 👉 Both sides use the **same formula** and **same classification**. The determinants differ because they reflect different underlying decisions (buying vs producing).

Summary

Today's Key Takeaways:

Demand elasticity recap:

1. $\epsilon_d = \frac{\Delta Q}{\Delta P}$ — negative, use $|\epsilon_d|$. Elastic > 1 , inelastic < 1
2. **Revenue test**: elastic $\rightarrow P \uparrow$ means $TR \downarrow$; inelastic $\rightarrow P \uparrow$ means $TR \uparrow$
3. **Determinants**: substitutes, budget share, necessity/luxury, time, market definition

Supply elasticity (new):

4. $\epsilon_s = \frac{\Delta Q_s}{\Delta P}$ — always positive. Same formula, same classification
5. **Linear supply through origin**: $\epsilon_s = 1$ everywhere. Positive intercept: $\epsilon_s > 1$
6. **Determinants**: flexibility, mobility, and replicability of inputs, **time horizon**
7. **Time**: very short run (inelastic) $\rightarrow$ short run $\rightarrow$ long run (elastic)
8. Both elasticities together determine **price volatility** and **tax burden sharing**

Next (Lecture 16, April 16): Short Run and Long Run Market Equilibrium 

Exercises

Practice Time! 

Demand and supply elasticity calculation.

Exercise 1: Multiple Choice

Question: The supply of hotel rooms in Porto's city center has $(\epsilon_s = 0.3)$. If room prices increase by 20%, what happens to the quantity of rooms supplied?

- A. Quantity increases by 60%
- B. Quantity increases by 6%
- C. Quantity increases by 0.6%
- D. Quantity increases by 20%

Answer: B

$$(\% \Delta Q_s = \epsilon_s \times \% \Delta P = 0.3 \times 20\% = 6\%)$$

Supply is **inelastic** ($(\epsilon_s = 0.3 < 1)$): even a large price increase (20%) produces only a small quantity response (6%). You can't build new hotel rooms quickly!

Exercise 2: Multiple Choice

Question: Demand for TAP flights from Lisbon to London has $(|\epsilon_d| = 1.8)$. If TAP raises fares by 10%, what happens to total revenue from this route?

- A. Revenue increases by 18%
- B. Revenue increases because higher price means more money per ticket
- C. Revenue falls because passenger numbers drop by more than 10%
- D. Revenue stays the same

Answer: C

Demand is **elastic** ($(|\epsilon_d| = 1.8 > 1)$): a 10% fare increase causes approximately $(1.8 \times 10\% = 18\%)$ fewer passengers. Since quantity drops **proportionally more** than price rises, $(TR = P \times Q)$ **falls**. This is the **revenue test** for elastic demand.

Exercise 3: Open Question

The market for *tuk-tuk* tours in Lisbon has the following estimated curves:

- **Demand:** $(P = 36 - 0.2Q)$ (€ per tour)
- **Supply:** $(P = 6 + 0.1Q)$ (€ per tour)

- Find the equilibrium price and quantity.
- Calculate the **price elasticity of demand** at the equilibrium point. Is demand elastic or inelastic?
- Calculate the **price elasticity of supply** at the equilibrium point. Is supply elastic or inelastic?
- At the equilibrium, if the city council imposes a small price increase (say €1), would you expect total revenue to rise or fall? Use the revenue test from part (b).
- The supply has a positive intercept ($(c = 6)$). Using the shortcut formula $(\epsilon_s = \frac{P}{P - c})$, verify your answer from part (c). At what price would supply elasticity equal exactly 2?
- The city council considers a €3 per-tour tax on operators. This shifts supply to $(P = 9 + 0.1Q)$. Find the new equilibrium. Who bears more of the tax burden — tourists or operators? Relate your answer to the elasticities you calculated.

Exercise 3: Solution — Parts a, b & c

a) Equilibrium: $(36 - 0.2Q = 6 + 0.1Q) \rightarrow (30 = 0.3Q) \rightarrow (Q^* = 100), (P^* = 36 - 0.2(100) = €16)$

b) Demand elasticity at $((Q^* = 100, P^* = 16))$:

Demand slope = (-0.2) , so $(\frac{1}{\text{slope}} = \frac{1}{-0.2} = -5)$

$(\epsilon_d = -5 \times \frac{16}{100} = -0.8)$

$(|\epsilon_d| = 0.8 < 1) \rightarrow$ Demand is **inelastic** at the equilibrium.

c) Supply elasticity at $((Q^* = 100, P^* = 16))$:

Supply slope = (0.1) , so $(\frac{1}{\text{slope}} = 10)$

$(\epsilon_s = 10 \times \frac{16}{100} = 1.6)$

$(\epsilon_s = 1.6 > 1) \rightarrow$ Supply is **elastic** at the equilibrium.

Exercise 3: Solution — Parts d & e

d) Since demand is **inelastic** ($|\epsilon_d| = 0.8 < 1$), the revenue test says: a price increase $\rightarrow$ total revenue **rises**. Quantity falls, but by a smaller percentage than price rises. The €1 increase would increase TR.

e) Shortcut: $\epsilon_s = \frac{P}{P - c} = \frac{16}{16 - 6} = \frac{16}{10} = 1.6$  Matches part (c)!

For $\epsilon_s = 2$: $\frac{P}{P - 6} = 2 \rightarrow P = 2P - 12 \rightarrow P = €12$

At $(P = 12)$: $(Q = \frac{12 - 6}{0.1} = 60)$ tours. So supply is more elastic at **lower prices** (closer to the intercept).

Exercise 3: Solution — Part f

f) New supply with €3 tax: $(P = 9 + 0.1Q)$

New equilibrium: $(36 - 0.2Q = 9 + 0.1Q) \rightarrow (27 = 0.3Q) \rightarrow (Q^* = 90)$, $(P^* = 36 - 0.2(90) = €18)$

	Before tax	After tax	Change
Price paid by tourists	€16	€18	+€2
Price received by operators (net of tax)	€16	$€18 - €3 = €15$	-€1
Quantity	100	90	-10

Of the €3 tax: tourists pay **€2 more**, operators receive **€1 less**.

👉 **Tourists bear more** (€2 out of €3) because **demand is more inelastic** ($(|\varepsilon_d| = 0.8)$) than supply is ($(\varepsilon_s = 1.6)$). The side with **less elastic** response absorbs more of the tax.



This is a **general rule**: tax burden falls more heavily on the side of the market that is **less responsive** to price changes.

Next Lecture

April 16, 2026: Short Run and Long Run Market Equilibrium 

We finally put **demand and supply together** to find prices and quantities!

👉 This is the payoff of Lectures 5–15: consumer theory meets producer theory!

Thank You!

Questions? 🙋

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Next class: Thursday, April 16, 2026